Reading list: LoRA rank and forgetting in small language models

Feasible Rank Window project · prepared 25 October 2026. Papers selected for direct relevance to rank × forgetting in sequential fine-tuning of sub-3B decoders. Foundational papers that only define methods or metrics (LoRA itself, BWT, EWC, catastrophic forgetting) are left out. Each entry states whether it is peer-reviewed or a preprint; preprints and the blog post should be cited with that caveat. Links were checked on 25 October 2026.

1. LoRA rank, learning and forgetting (core to the research question)

Direct evidence on how LoRA rank relates to learning and forgetting.

1. MoRA: High-Rank Updating for Parameter-Efficient Fine-Tuning
Jiang et al. · **arXiv preprint, 2024** · Preprint
https://arxiv.org/abs/2405.12130
Why it matters: Argues low-rank updates limit learning of new knowledge and gives a high-rank alternative at equal parameter count; the opposite position to the flat rank curve in v4, useful for the discussion.

2. LoRA Training in the NTK Regime has No Spurious Local Minima
Jang, Lee, Ryu · **ICML, 2024 (oral)** · Peer-reviewed
https://arxiv.org/abs/2402.11867
Why it matters: Proves LoRA with rank around the square root of the number of training examples avoids bad local minima on NLP classification; a theoretical lower bound on the rank needed, relevant to why small ranks suffice.

3. Revisiting Catastrophic Forgetting in Large Language Model Tuning
Li, Ding, Fang, Tao · **Findings of EMNLP, 2024** · Peer-reviewed
https://aclanthology.org/2024.findings-emnlp.249/
Why it matters: Links forgetting in LLM fine-tuning to the sharpness of the loss landscape; a mechanism for why the learning rate, not rank, drove forgetting in v4.

4. An Empirical Investigation of the Role of Pre-training in Lifelong Learning
Mehta, Patil, Chandar, Strubell · **JMLR 24(214), 2023** · Peer-reviewed
https://jmlr.org/papers/v24/22-0496.html
Why it matters: Pre-trained models forget less in task sequences (15 NLP tasks) because they sit in wider minima; background for the model-size and family differences in v4.

5. Scaling Laws for Forgetting When Fine-Tuning Large Language Models
Kalajdzievski · **arXiv preprint, 2024** · Preprint
https://arxiv.org/abs/2401.05605
Why it matters: Forgetting follows a power law in the number of fine-tuned parameters and steps, and LoRA still forgets: a quantitative frame for rank as 'parameters fine-tuned'.

6. Adaptive Rank, Reduced Forgetting: Continual Learning with Dynamic Rank-Selective LoRA (CoDyRA)
Lu, Zhao, Xue, Yao, Moore, Gong · **arXiv preprint, 2024–2026** · Preprint
https://arxiv.org/abs/2412.01004
Why it matters: Higher rank gives more plasticity and more forgetting in continual learning; motivates rank as a stability–plasticity lever.

7. Subspace Geometry Governs Catastrophic Forgetting in Low-Rank Adaptation
Steele · **arXiv preprint, 2026** · Preprint (single author)
https://arxiv.org/abs/2603.02224
Why it matters: Argues forgetting is set by the angle between task-gradient subspaces and is nearly rank-invariant: one explanation for the flat BWT-vs-rank curve in v4.

8. Understanding Catastrophic Forgetting in LoRA via Mean-Field Attention Dynamics
(2024 authors) · **arXiv preprint, 2024** · Preprint
https://arxiv.org/abs/2402.15415
Why it matters: A theoretical account of how LoRA updates to attention cause forgetting; relevant because v4 adapts q/v attention matrices.

9. On Catastrophic Forgetting in Low-Rank Decomposition-Based Parameter-Efficient Fine-Tuning
(2026 authors) · **arXiv preprint, 2026** · Preprint
https://arxiv.org/abs/2603.09684
Why it matters: Compares LoRA-family methods under sequential training and finds low ranks forget most; a counterpoint to discuss (vision setting).

10. Rethinking the Rank Threshold for LoRA Fine-Tuning
Park · **arXiv preprint, 2026** · Preprint (single author)
https://arxiv.org/abs/2605.03724
Why it matters: Theory and experiments suggesting r = 1 suffices for binary GLUE tasks, while MNLI needs more; matches the v4 observation that r = 1 already learns these tasks.

2. Learning rate and the LoRA parameterisation (why v3 was confounded)

Papers that explain why rank cannot be compared without controlling the learning rate.

11. Learning Rate Scaling across LoRA Ranks and Transfer to Full Finetuning (μA)
Chen, Villar, Hayou · **ICML (PMLR v306), 2026** · Peer-reviewed
https://arxiv.org/abs/2602.06204
Why it matters: Shows the optimal LR is roughly rank-invariant with α/r scaling but shrinks with rank under a constant multiplier (the v3 $\alpha = 2r$ setting). The key citation for the v3 $\rightarrow$ v4 change.

12. A Rank Stabilization Scaling Factor for Fine-Tuning with LoRA (rsLoRA)
Kalajdzievski · **arXiv preprint, 2023** · Preprint
https://arxiv.org/abs/2312.03732
Why it matters: Proposes $\alpha/\sqrt{r}$ so high ranks do not slow learning; an alternative parameterisation to discuss.

13. LoRA+: Efficient Low Rank Adaptation of Large Models
Hayou, Ghosh, Yu · **ICML, 2024** · Peer-reviewed
https://arxiv.org/abs/2402.12354
Why it matters: Different LRs for the A and B factors; relevant to how LoRA's optimisation depends on its parameterisation.

14. LoRA Without Regret
Schulman and Thinking Machines Lab · **Technical blog, 2025** · Not peer-reviewed
https://thinkingmachines.ai/blog/lora/
Why it matters: With α fixed, the optimal LR varies under 2× from rank 4 to 512; all-linear LoRA beats attention-only. Supports both the v4 parameterisation and the all-linear decision.

15. Simple and Scalable Strategies to Continually Pre-train Large Language Models
Ibrahim, Thérien, Gupta, et al. · **TMLR, 2024** · Peer-reviewed
https://arxiv.org/abs/2403.08763
Why it matters: Learning-rate re-warming at each new stage causes forgetting, and replay plus schedule changes reduce it; directly relevant to the per-task LR restart proposed for the next phase.

16. Learning Rate Matters: Vanilla LoRA May Suffice for LLM Fine-tuning
Lee, Ko, Chen, Yeh · **arXiv preprint, 2026** · Preprint
https://arxiv.org/abs/2602.04998
Why it matters: Comparisons between LoRA configurations can flip once the LR is tuned; justifies the per-rank LR search.

3. Model scale and forgetting (the sub-3B question)

Evidence on how model size changes forgetting, relevant to RQ2/RQ4.

17. An Empirical Study of Catastrophic Forgetting in Large Language Models During Continual Fine-tuning
Luo, Yang, Meng, Li, Zhou, Zhang · **arXiv preprint, 2023 (revised 2025)** · Preprint
https://arxiv.org/abs/2308.08747
Why it matters: Forgetting measured across 1B–7B decoders; reports more severe forgetting at larger scale, a useful contrast to the Qwen 1.5B $\rightarrow$ 3B result in v4.

18. Scaling Laws for Forgetting during Finetuning with Pretraining Data Injection
Bethune et al. · **arXiv preprint, 2025** · Preprint
https://arxiv.org/abs/2502.06042
Why it matters: Forgetting scaling laws across several model sizes, and how mixing in pretraining data reduces it; relevant to size effects and to cheap mitigations.

19. Effect of Scale on Catastrophic Forgetting in Neural Networks
Ramasesh, Lewkowycz, Dyer · **ICLR, 2022** · Peer-reviewed
https://openreview.net/pdf/54d452543db390a26979d51399ae9c6ed01f4de1.pdf
Why it matters: Larger pretrained models forget less, linked to more orthogonal representations; theory for why Qwen2.5-3B forgot least.

20. The Importance of Being Lazy: Scaling Limits of Continual Learning
(ICML 2025 authors) · **ICML, 2025** · Peer-reviewed
https://arxiv.org/abs/2506.16884
Why it matters: Whether scale reduces forgetting depends on the parameterisation (it does under NTP, not under μP); relevant to interpreting size effects.

4. What 'forgetting' really is (supports the probe finding)

Papers showing that measured forgetting is often misalignment, not lost knowledge.

21. Spurious Forgetting in Continual Learning of Language Models
Zheng, Cai, Qiu, Ma · **ICLR, 2025** · Peer-reviewed
https://arxiv.org/abs/2501.13453
Why it matters: Much apparent forgetting is lost task alignment rather than lost knowledge, which matches the v4 linear-probe result (head forgetting ■ representation forgetting).

22. Probing Representation Forgetting in Supervised and Unsupervised Continual Learning
Davari, Asadi, Mudur, Aljundi, Belilovsky · **CVPR, 2022** · Peer-reviewed
https://openaccess.thecvf.com/content/CVPR2022/html/Davari_Probing_Representation_Forgetting_in_Supervised_and_Unsupervised_Continual_Learning_CVPR_2022_paper.html
Why it matters: The linear-probe method used in v4 to separate representation forgetting from head misalignment.

23. Fine-Tuning can Distort Pretrained Features and Underperform Out-of-Distribution (LP-FT)
Kumar, Raghunathan, Jones, Ma, Liang · **ICLR, 2022** · Peer-reviewed
https://arxiv.org/abs/2202.10054
Why it matters: Training a randomly initialised head distorts features; LP-FT is the proposed fix for the head policy in the next phase.

24. Intrinsic Dimensionality Explains the Effectiveness of Language Model Fine-Tuning
Aghajanyan, Zettlemoyer, Gupta · **ACL, 2021** · Peer-reviewed
https://aclanthology.org/2021.acl-long.568/
Why it matters: GLUE-type tasks need very few parameters, and fewer for larger models; explains why rank barely affects accuracy on RTE/MNLI/SST-2.

5. LoRA-based continual learning of LLMs (methods and baselines)

Methods your results can be compared with or extended towards.

25. SAPT: A Shared Attention Framework for Parameter-Efficient Continual Learning of LLMs
Zhao et al. · **ACL, 2024** · Peer-reviewed
https://aclanthology.org/2024.acl-long.625/
Why it matters: Parameter-efficient continual learning across 770M–13B models; a strong baseline for longer task sequences.

26. InfLoRA: Interference-Free Low-Rank Adaptation for Continual Learning
Liang, Li · **CVPR, 2024** · Peer-reviewed
https://arxiv.org/abs/2404.00228
Why it matters: Constrains each task's LoRA update to a subspace that does not interfere with earlier tasks; a principled single-adapter alternative to compare with plain sequential LoRA.

27. Merge before Forget: A Single LoRA Continual Learning via Continual Merging (SLAO)
Qiao, Mahdavi · **ICLR, 2026** · Peer-reviewed
https://arxiv.org/abs/2512.23017
Why it matters: Keeps one LoRA across tasks (like v4) but merges updates with orthogonal initialisation; tested on Llama models with constant memory.

28. Mitigating Catastrophic Forgetting in Large Language Models with Self-Synthesized Rehearsal
Huang et al. · **ACL, 2024** · Peer-reviewed
https://aclanthology.org/2024.acl-long.77/
Why it matters: Replay without the original data, using model-generated examples; a cheap mitigation baseline for the next phase.

29. Continual Learning with Low Rank Adaptation (CoLoR)
Wistuba, Sivaprasad, Balles, Zappella · **NeurIPS Workshop on Distribution Shifts, 2023** · Peer-reviewed (workshop)
https://arxiv.org/abs/2311.17601
Why it matters: Shows LoRA-based continual learning beats prompt-tuning methods at similar parameter cost.

30. Analyzing and Reducing Catastrophic Forgetting in Parameter Efficient Tuning (I-LoRA)
Ren, Li, Wang, Zhao, Qin · **arXiv preprint, 2024** · Preprint
https://arxiv.org/abs/2402.18865
Why it matters: Mode connectivity in LLM continual learning with LoRA; interpolation-based method to balance stability and plasticity.

31. Fine-tuned Language Models are Continual Learners
Scialom, Chakrabarty, Muresan · **EMNLP, 2022** · Peer-reviewed
https://aclanthology.org/2022.emnlp-main.410/
Why it matters: Sequential learning of 8 generation tasks with retention; relevant to the move to generative tasks.

32. TRACE: A Comprehensive Benchmark for Continual Learning in Large Language Models
Wang et al. · **arXiv preprint, 2023** · Preprint
https://arxiv.org/abs/2310.06762
Why it matters: A ready-made mixed-task sequence for LLM continual learning, including general-ability retention checks.

33. Hyperparameters in Continual Learning: A Reality Check
Cha, Cho · **TMLR, 2025** · Peer-reviewed
https://arxiv.org/abs/2403.09066
Why it matters: Tuning on the evaluated stream inflates results; supports the separate tuning seeds and training holdout used in v4.

34. Continual Learning of Large Language Models: A Comprehensive Survey
Shi et al. · **ACM Computing Surveys, 2025** · Peer-reviewed (survey)
https://arxiv.org/abs/2404.16789
Why it matters: Map of the field to position the project and find further related work.

34 papers. Rebuild with: .venv/bin/python tools/build_reading_list_pdf.py